Original-Drug Innovation, Implementation-Route Assignment, and the MAH System in China

Mechanism, Dynamic Model, and Empirical Strategy

April 29, 2026

1. Main Question and Novelty

Research question

Does MAH raise **realized original-drug innovation** by changing implementation routes for research-originated ideas?

- MAH is modeled as a change in **implementation-route assignment**, not as a generic innovation shock.
- The mechanism is embedded in a **dynamic industry model** with switching, entry and exit, and a qualified contract-manufacturing market.

What is new here relative to adjacent literatures?

- Not a generic innovation-stimulus paper: the reform acts on **implementation-route feasibility**, not directly on scientific productivity.
- Not a pure regulation or productivity paper: the main outcome is **realized original-drug products**, not factory-level efficiency alone.
- Not a standard outsourcing paper: the key object is **authorization-retaining external implementation** under MAH.

2. Before MAH: Why Implementation Is the Bottleneck

- In pharmaceuticals, invention and product realization need not occur inside the same firm.
- Before MAH, in China, **authorization holding was tightly tied to production qualification**. That linkage made external implementation much harder for research-originating firms.
- A research-originating firm with an original-drug idea often faced three costly options:
  1. build or acquire internal manufacturing capacity;
  2. transfer effective control to an integrated producer;
  3. leave the project unrealized or delayed.
- The bottleneck is therefore often **organizational implementation**, not only scientific discovery.
- This is why the paper focuses on **original-drug realization** rather than on total pharmaceutical output alone.

Core intuition: MAH matters if it changes whether a research-originated original-drug idea can reach the market while the innovator retains authorization and project control.

3. Institutional Evolution under MAH

- **2016 pilot:** selected provinces began to allow MAH-style decoupling between authorization holding and in-house production.
- **2019 Drug Administration Law:** a holder may self-manufacture or entrust a **qualified** outside manufacturer; entrusted production requires formal entrustment and quality agreements.
- **2022/2023 implementing rules:** holder responsibility, quality management, and regulatory supervision were clarified and strengthened.

Institutional interpretation

MAH does not simply permit generic outsourcing. It creates a legally structured **external implementation route** under which the innovator may retain authorization while production is carried out by a qualified outside manufacturer. In the model, that is a change in **authorization–production assignment**.

4. Route-Specific Policy Primitive

Policy primitive

$$\tau_{ext}(1) < \tau_{ext}(0),$$

where $M = 0$ is the pre-MAH regime and $M = 1$ is the post-MAH regime.

- $\tau_{ext}(M)$: route-specific governance and compliance friction of external implementation.
- The innovator retains authorization status while relying on a qualified external manufacturer.
- τ_{ext} is **not** scientific productivity, **not** the physical manufacturing price p_m, and **not** a generic policy dummy.
- MAH enters the model through τ_{ext}, while implementation success, manufacturing prices, and alternative routes remain distinct objects.
- The model then adds heterogeneous firms, organizational switching, entry and exit, and a qualified contract-manufacturing market around that primitive policy margin.

$$\text{MAH} \Rightarrow \tau_{ext} \downarrow \Rightarrow \text{external implementation value} \uparrow \Rightarrow \text{realized original-drug innovation} \uparrow .$$

5. Baseline Environment

The baseline is a **lower-dimensional dynamic industry model**. It keeps only the objects needed to connect MAH to original-drug innovation through implementation-route assignment.

Object	**Role in the model**
Policy regime $M \in \{0, 1\}$	Pre-MAH versus post-MAH institutional environment
Capability $z \in Z$	One-dimensional firm heterogeneity
Type $s \in \{A, B, C\}$	Organizational form and implementation role
Friction $\tau_{ext}(M)$	How MAH enters the model
Manufacturing price p_m	Price of qualified external production services
Factor price w	Static operating-cost shifter in current profits

Discipline

The baseline includes route assignment, organizational switching, entry and exit, and a qualified contract-manufacturing market. It does **not** build a full incomplete-contract model, a full bargaining model, or a government-optimization problem.

6. Timing Within a Period

For each period t, the sequence is:

1. **Institutional regime:** firms observe M_t and the relevant external manufacturing price $p_{m,t}$.
2. **State:** each incumbent observes its organizational state $s_t \in \{A, B, C\}$ and capability z_t.
3. **Research effort:** type-A chooses $(x^G_{A,t}, x^O_{A,t})$; type-B chooses $x^O_{B,t}$ only. Here x denotes innovation intensity, not an integer project count.
4. **Idea arrival:** innovation intensity generates generic or original-drug project arrivals probabilistically.
5. **Implementation assignment:** original-drug ideas are assigned across feasible routes $H = \{ext, int, tr, ab\}$.
6. **Organization choice:** incumbents may stay, switch organizational form, or exit.
7. **Continuation and entry:** next-period states, entry values, and market-clearing objects are determined.

Where MAH matters in timing

MAH acts directly at stage 5 through $\tau_{ext}(M_t)$, and then feeds back to stages 3, 6, and 7 through the value of original-drug projects, continuation values, and entry incentives.

7. Economic Roles of the Three Types

Type A

Integrated firms

- choose x_A^G, x_A^O
- can implement internally
- provide the within-firm direction margin

Type B

Research-specialized firms

- choose x_B^O only
- assign original-drug ideas across routes
- are most exposed to τ_{ext}

Type C

Manufacturing specialists

- provide qualified production
- support the external route
- help determine p_m

Interpretation: A/B/C are economic organizational forms that govern who invents, who implements, and which routes are feasible for the marginal project.

8. Type-B Innovation Intensity and Route Assignment

Type-B firms are the central mechanism objects in the paper. In the baseline they are research-specialized entrants or organizations whose main economically relevant margin is original-drug innovation, so they do **not** choose generic-project effort.

Type-B payoff

$$\Pi_B(z; M, p_m) = -f_B + \max_{x_B^O \geq 0} \left\{ x_B^O \Psi_B^O(z; p_m, M) - C_B(x_B^O; z) \right\}.$$

- x_B^O: type-B original-drug innovation intensity, interpretable as idea-arrival intensity or effort devoted to generating original-drug projects; it is not an integer count of ideas.
- $\Psi_B^O(z; p_m, M)$: expected value of one type-B original-drug idea after optimal route assignment.
- Thus $x_B^O \Psi_B^O$ is expected flow value: innovation intensity times per-idea expected value.
- For each original-drug idea, the feasible route set is $H = \{ext, int, tr, ab\}$.

Key margin: MAH raises the value of the external route, which raises Ψ_B^O, and therefore lifts original-drug innovation intensity, continuation value, entry value, and realization.

9. Type-A Internal Benchmark

Type-A firms are integrated firms. They can implement internally, so they are not the central MAH-specific actor, but they help explain industry-level project composition.

$$\Pi_A(z; M, w, p_m) = \bar{\pi}_A(z; w, r, c_m) + \max_{x_A^G, x_A^O \geq 0} \left\{ x_A^G \Psi_A^G(z) + x_A^O \Psi_A^O(z; M) - C_A(x_A^G, x_A^O; z) \right\}.$$

- x_A^G: generic-oriented effort by integrated firms.
- x_A^O: original-drug-oriented effort by integrated firms.
- This block captures within-type-A generic-to-original portfolio adjustment.

Role in the paper

Type-A provides the internal benchmark. It helps explain within-firm direction adjustment without forcing the main MAH-specific mechanism into type-B generic/original switching.

10. Type-C and the Qualified Manufacturing Market

Type-C firms are not a second innovation actor in the baseline. Their role is to supply the qualified manufacturing services needed for external implementation.

$$\Pi_C(z; p_m, w) = \bar{\pi}_C(z; p_m, w).$$

- Type-C supplies qualified production capacity.
- The external-route price p_m summarizes the market cost of using those services.
- The external route creates a market link between type-B demand and type-C supply:

$$S_m(p_m, w; M) = D_m(p_m; M).$$

- Producer-side entry, profits, and general-equilibrium feedbacks are supporting margins, not the final innovation outcome.

Why this simplification helps

It keeps the main model focused on how MAH changes original-drug implementation for type-B research firms, while still allowing the qualified-manufacturing price p_m to matter in equilibrium.

11. Original-Drug Route Assignment

For a type-B original-drug idea, the per-idea expected value is the best feasible route:

$$\Psi_B^O(z; p_m, M) = \max\left\{H_B^{ext}(z, p_m; M),\ H_B^{int}(z),\ H_B^{tr}(z),\ 0\right\}.$$

The external route value is

$$H_B^{ext}(z, p_m; M) = \lambda_{ext}(z, p_m) R_B(z) - p_m - \tau_{ext}(M).$$

- *ext*: retain authorization and use qualified external production.
- *int*, *tr*, and *ab*: internal implementation, transfer, or abandonment.
- MAH directly shifts only the external route by lowering $\tau_{ext}(M)$.

Simple decision rule

The external route is chosen when H_B^{ext} dominates the other feasible routes.

12. Comparative-Static Logic

One-sentence mechanism

When MAH lowers τ_{ext}, the external route becomes more attractive for type-B original-drug ideas, which raises per-idea value, research effort, and realization.

1. **Route margin:** the external-route choice set expands.
2. **Innovation-intensity margin:** higher per-idea value Ψ_B^O raises x_B^O.
3. **Dynamic margin:** higher project value raises continuation and entry value.
4. **Conversion margin:** more original-drug ideas become realized original-drug products.

$$\tau_{ext} \downarrow \Rightarrow H_B^{ext} \uparrow \Rightarrow \Psi_B^O \uparrow \Rightarrow x_B^O,\ V_B,\ \mathrm{Conv}_B^O \uparrow .$$

Caveat: type-B entry and type-C profits are equilibrium objects, not unconditional monotone outcomes.

13. First-Layer Objects and Final Outcomes

The first empirical objects should be the ones closest to the mechanism itself:

- external-route use or holder–producer separation;
- realized original-drug products;
- the share of original drugs among realized products;
- research-oriented entry and continuation.

The same logic can be summarized through three final innovation outcomes:

$$N_O(M), \qquad S_O(M) = \frac{N_O(M)}{N_O(M) + N_G(M)}, \qquad \mathrm{Conv}_O(M).$$

- $N_O(M)$: number of realized original-drug products.
- $S_O(M)$: original-drug share among realized products.
- $\mathrm{Conv}_O(M)$: conversion from original-drug idea/application to realized product.

14. Interpreting the Original-Drug Share

Under the revised baseline, a rise in S_O is not interpreted as type-B switching from generic to original projects.

$$\Delta S_O = \underbrace{\Delta S_O^{A,within}}_{\text{type-A G/O portfolio adjustment}} + \underbrace{\Delta S_O^{B,entry}}_{\text{more original-drug-oriented B entry}} + \underbrace{\Delta S_O^{B,conversion}}_{\text{higher B original-drug realization}} .$$

- Type-A explains within-firm direction adjustment.
- Type-B explains original-drug entry and conversion.
- Type-C supplies the qualified production capacity needed for the external route.

Empirical discipline

This decomposition is most persuasive if the data support two diagnostics: pre-reform research-oriented firms or entities with weak internal manufacturing capacity are more exposed to the implementation bottleneck; and integrated type-A firms display meaningful generic/original portfolio variation. The *B*-entry term should be interpreted as post-reform emergence or expansion of research-specialized original-drug organizations, not as evidence that such MAH-style type-B firms already existed before the reform.

15. Dynamic Industry Block

The static route mechanism feeds into dynamic organizational choice, continuation values, and entry.

Bellman logic

$$V_s(z; M) = \max\left\{0, \max_{j \in \mathcal{J}(s)} W_{sj}(z; M)\right\}, \qquad s \in \{A, B, C\}.$$

- Lower τ_{ext} raises the current-period value of the type-B external route and therefore Π_B.
- Higher current payoff lifts continuation value and changes stay/switch/exit choices.
- Higher V_B raises the value of entering as a type-B research firm.

Caveat: the model most cleanly predicts a higher type-B *entry value*; an observed increase in entry mass requires entry-cost heterogeneity or an elastic supply of potential entrants.

16. Empirical Mapping: First-Layer and Second-Layer Objects

Model object	Empirical object	Interpretation
$\tau_{ext} \downarrow$	Holder–producer separation; external-route use	External implementation becomes more feasible
$H_B^{ext} \uparrow$	More MAH-style external implementation under retained authorization	Route assignment changes
$\Psi_B^O \uparrow$	Type-B entry and continuation	Original-drug project value rises
$V_B \uparrow$	Type-B survival and entry	Dynamic response to higher project value
$x_B^O \uparrow$	Original-drug project starts, applications, R&D effort, or idea-arrival proxies	Research-specialized firms raise original-drug innovation intensity
$\mathrm{Conv}_B^O \uparrow$	Application-to-approval or idea-to-product conversion	Realization becomes more feasible
$N_O, S_O \uparrow$	Realized original-drug products and share	Final innovation outcomes

Hierarchy

First-layer objects are route use, realization, and original-drug outcomes. Transitions and type-C outcomes are second-layer validation objects.

17. Data, Exposure Construction, and Type Outcomes

Pre-reform exposure proxies

Use **pre-reform** characteristics to measure exposure to the implementation bottleneck, not to claim that MAH-style type-B firms already existed before the reform.

- Integrated-incumbent exposure: original-drug activity plus relevant internal manufacturing capability.
- Potential type-B exposure: research-oriented firms or entities with weak or no internal manufacturing capability.
- Type-C exposure: qualified manufacturing specialists that can support external implementation.

Exposure design

The main treatment margin is whether MAH matters more for firms or regions that were more constrained by the pre-reform implementation bottleneck.

- Baseline exposure: pre-reform research orientation combined with weak internal manufacturing capability, rather than post-reform MAH-holder status.
- Baseline heterogeneity margins: pre-reform R&D capacity and pre-reform manufacturing capability.
- Post-reform R&D effort is treated as a mechanism outcome, not as a default control variable.

18. Identification, Estimation, and Current Claim Boundary

Identification logic

The data should first ask whether MAH changes holder–producer separation, external-route use, realized original-drug products, and responses among firms more exposed to pre-reform implementation bottlenecks.

Empirical strategy

The current design is two-step: reduced-form evidence on first-layer objects first; then a parsimonious stationary model, disciplined by those patterns, for later SMM or minimum-distance estimation.

What this draft does and does not claim

The draft argues that MAH changes implementation-route assignment rather than scientific productivity directly. It does **not** claim to observe τ_{ext} directly, to infer the mechanism from innovation outcomes alone, or to treat organizational transitions as identifying by themselves.

19. Current Model-Data Gaps

What the current data can already discipline

- holder–producer separation or other route-use evidence;
- realized original-drug products and original-drug shares;
- exposure heterogeneity across firms or regions with tighter pre-reform implementation constraints.

What still needs stronger measurement or extra structure

- a cleaner project-to-route measure for the full assignment problem;
- application-to-product linkage for a sharper conversion object Conv_O;
- stronger producer-side data for the qualified manufacturing market and p_m-side validation;
- richer transition and entry objects if we want a full stationary SMM exercise rather than a reduced-form-plus-discipline first paper.

20. Questions

What we want to confirm

1. Is the main question sufficiently new and important: MAH as a change in implementation-route assignment, rather than as a generic innovation shock?
2. Is the current baseline disciplined enough: type-A with x_A^G, x_A^O, type-B with x_B^O, and type-C supplying the qualified manufacturing market?
3. Is the empirical hierarchy right: route use and realization first, organizational transitions and producer-side outcomes second?
4. For the first full paper version, should we stop at reduced-form evidence plus a disciplined mechanism model, or move directly to a fuller stationary estimation block?